

Best configuration per algorithm and flow ordering (SFT size = 50). Reactive baseline hit rate: 67.87%. Each row reports the aging factor that maximizes the reported metric.

(a) Speculative Hit Rate

RL	Order	Aging Factor	Hit Rate (%)	Δ Reactive
Bandit	Source	0.999	49.42	-18.45
Bandit	Destination	0.999	48.42	-19.45
Bandit	Trace	0.997	50.28	-17.59
PPO	Source	0.999	20.40	-47.47
PPO	Destination	0.997	29.18	-38.69
PPO	Trace	0.997	31.05	-36.82
DQN	Source	0.999	20.73	-47.14
DQN	Destination	0.999	29.44	-38.43
DQN	Trace	0.999	31.42	-36.45

(b) Speculative+Reactive Hit Rate

RL	Order	Aging Factor	Hit Rate (%)	Δ Reactive
Bandit	Source	0.993	69.35	+1.48
Bandit	Destination	0.993	69.10	+1.23
Bandit	Trace	0.9	80.82	+12.95
PPO	Source	0.993	68.67	+0.80
PPO	Destination	0.993	68.86	+0.99
PPO	Trace	0.9	75.03	+7.16
DQN	Source	0.993	68.55	+0.68
DQN	Destination	0.993	68.74	+0.87
DQN	Trace	0.95	74.70	+6.83

(c) Speculative+Reactive Speculation Efficiency

RL	Order	Aging Factor	Speculation Efficiency
Bandit	Source	0.993	11.82
Bandit	Destination	0.997	11.77
Bandit	Trace	0.993	11.55
PPO	Source	0.997	10.73
PPO	Destination	0.997	10.55
PPO	Trace	0.995	7.62
DQN	Source	0.997	11.97
DQN	Destination	0.993	9.80
DQN	Trace	0.991	7.21